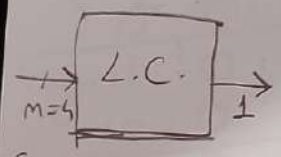
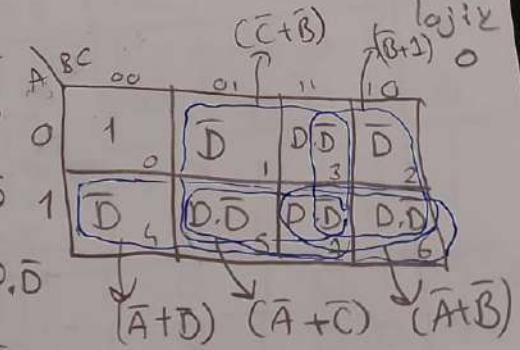
Ör: Bir hastanede 5 tane asansör vardır. 5. asansör yedek olarak kullanılıyor. 4 asansörden 2 veya daha fazlası meşgul olduğunda 5. asansör devreye alınacaktır. Böyle bir sisteme ait lojik devreyi sadece NOR kapılarıyla gerçekleştiriniz. 3. Mertebeden K. map kullanabilirsiniz.



K. map
LSB

	Dec.	A	B	C	D	f	
0	0	0	0	0	0	1	1
	1	0	0	0	1	1	
1	2	0	0	1	0	1	$\bar{D}$
	3	0	0	1	1	0	
2	4	0	1	0	0	1	$\bar{D}$
	5	0	1	0	1	0	
3	6	0	1	1	0	0	$D.\bar{D}$
	7	0	1	1	1	0	
4	8	1	0	0	0		$\bar{D}$
	9	1	0	0	1	0	
5	10	1	0	1	0	0	$D\bar{D}$
	11	1	0	1	1	0	
6	12	1	1	0	0	0	$D.\bar{D}$
	13	1	1	0	1	0	
7	14	1	1	1	0	0	$D.\bar{D}$
	15	1	1	1	1	0	



$N-n = 2^1 = 2$

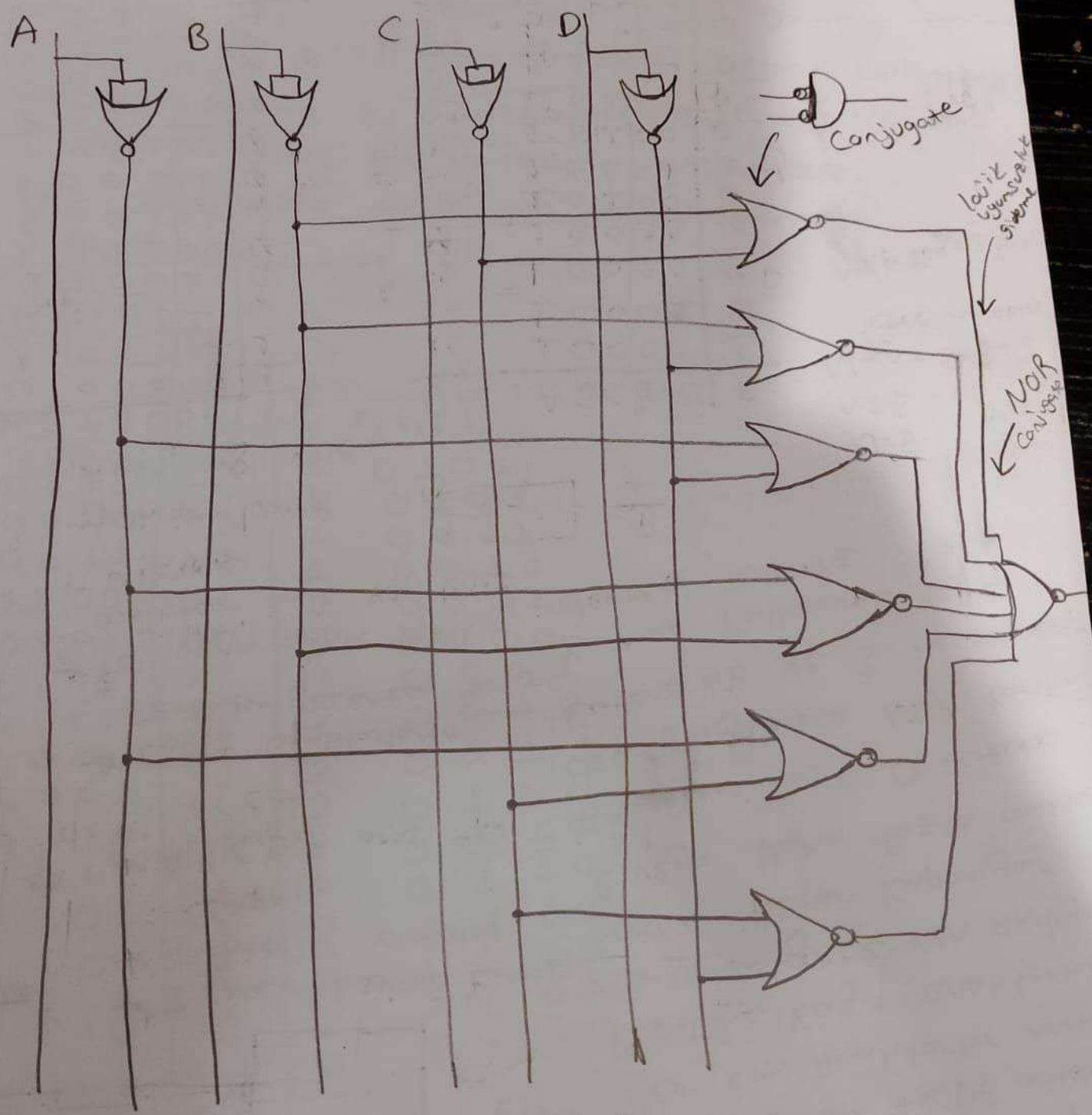
$N=4$

$n=3$

$N-n = 4-3 = 1$ free variable

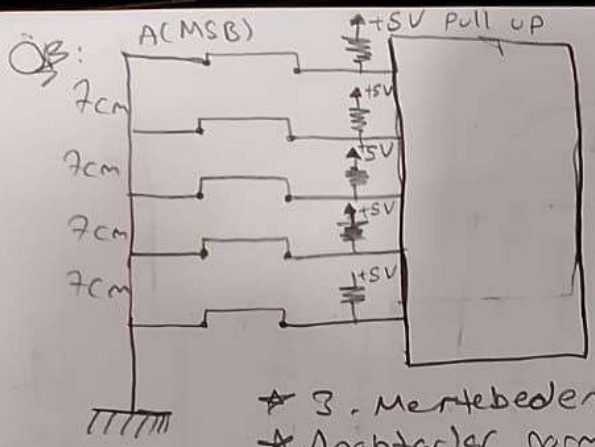
$$\overline{f_{POS}} = \frac{(\bar{B}+\bar{C}) \cdot (\bar{B}+\bar{D}) \cdot (\bar{A}+\bar{D}) \cdot (\bar{A}+\bar{B}) \cdot (\bar{A}+\bar{C}) \cdot (\bar{C}+\bar{D})}{(\bar{B}+\bar{C}) + (\bar{B}+\bar{D}) + (\bar{A}+\bar{D}) + (\bar{A}+\bar{B}) + (\bar{A}+\bar{C}) + (\bar{C}+\bar{D})}$$

NOR $\overline{A+B}$ $\overline{X+y}$



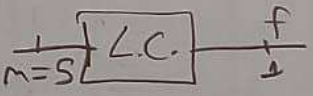
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Şekilde bir fotokobi makinesi görülüyor. ABCDE fotokobi makinesinde kağıdın yolu üzerine sırayla dizilmiş anahtarlardır. ABCDE arası 7'er cm. Anahtarlar normalde kapalıdır. Kağıt anahtarların üzerinden geçerken açılıyor sonra tekrar kapanıyor. 2 veya daha fazla anahtar açık iken lojik 0 verecek devreyi yalnızca NOR kapıları kullanarak ve 3. mertebeden K. map kullanarak lojik devreyi çiziniz.





★ 3. Mertebeden K.map
★ Anahtarlar normalde kapalı
★ A,B,C,D,E arası 7'er cm
★ Kağıt anahtarların üzerinden geçerken açılıyor.
★ İki veya daha fazla anahtar açık iken lojik 0 çıkışı
Yalnızca NOR
★ min maliyetle

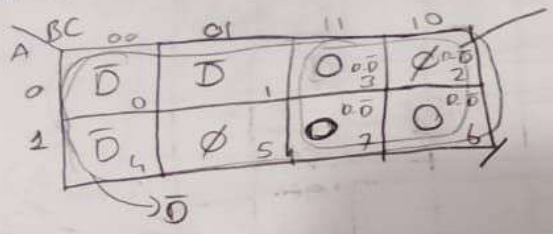


N=5
n=3
N−n=5−3=2 free variables

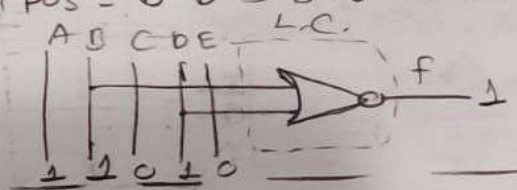
Map key = $2^3=4$

Dec	ABC	DE	f	
0	000	00	1	
	000	01	1	$\bar{D}$
	000	10	∅	
	000	11	0	
1	001	00	∅	
	001	01	∅	$\bar{D}$
	001	10	∅	
	001	11	∅	
2	010	00	∅	
	010	01	∅	∅
	010	10	∅	
	010	11	∅	
3	011	00	∅	
	011	01	∅	0
	011	10	0	
	011	11	0	

Dec	ABC	DE	f	
4	100	00	1	
	100	01	∅	$\bar{D}$
	100	10	∅	
	100	11	∅	
5	101	00	∅	
	101	01	∅	∅
	101	10	∅	
	101	11	∅	
6	110	00	0	
	110	01	∅	0
	110	10	∅	
	110	11	∅	
7	111	00	0	
	111	01	∅	0
	111	10	0	
	111	11	∅	

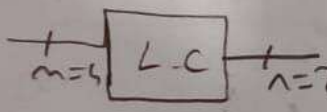


$f_{pos} = \overline{B} \cdot \overline{D} = \overline{\overline{B} \cdot \overline{D}} = \overline{B+D}$



ÖR: $f = x - 5$ fonksiyonunu yalnızca NAND kapılarını kullanarak gerçekleştiriniz. Burada x, 4 bitlik bir değişkendir. Negatif Negatif değerler kırmızı bir lamba ile gösterilecek. Fonksiyonu gerçekleştiren devreyi min maliyetle kurunuz.

Çıkış: Sign + Magnitude olsun.



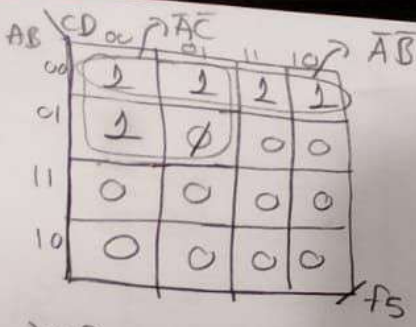
Dec.	A B C D	$f = x-5$	sign: f_s	M $f_3\ f_2\ f_1\ f_0$
0	0 0 0 0	-5	1	0 1 0 1
1	0 0 0 1	-4	1	0 1 0 0
2	0 0 1 0	-3	1	0 0 1 1
3	0 0 1 1	-2	1	0 0 1 0
4	0 1 0 0	-1	1	0 0 0 1
5	0 1 0 1	0	$\emptyset$	0 0 0 0
6	0 1 1 0	+1	0	0 0 0 1
7	0 1 1 1	+2	0	0 0 1 0
8	1 0 0 0	+3	0	0 0 1 1
9	1 0 0 1	+4	0	0 1 0 0
10	1 0 1 0	+5	0	0 1 0 1
11	1 0 1 1	+6	0	0 1 1 0
12	1 1 0 0	+7	0	0 1 1 1
13	1 1 0 1	+8	0	1 0 0 0
14	1 1 1 0	+9	0	1 0 0 1
15	1 1 1 1	+10	0	1 0 1 0

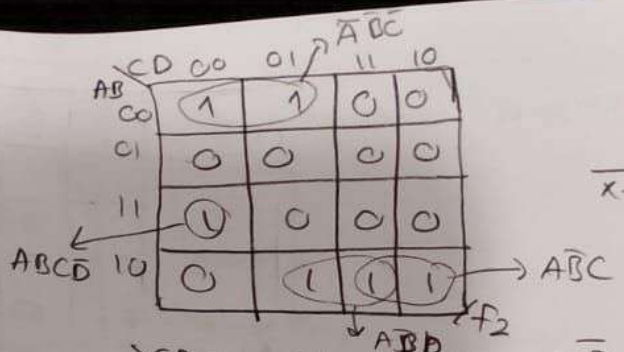
Sign

+5 → 0 101

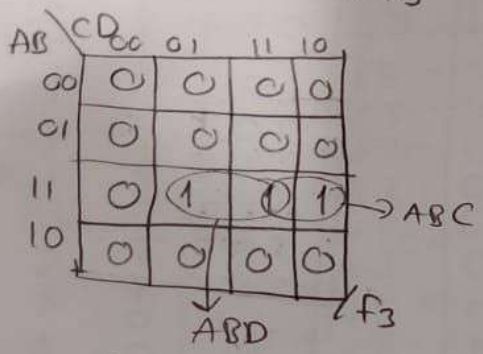
-5 → 1 101

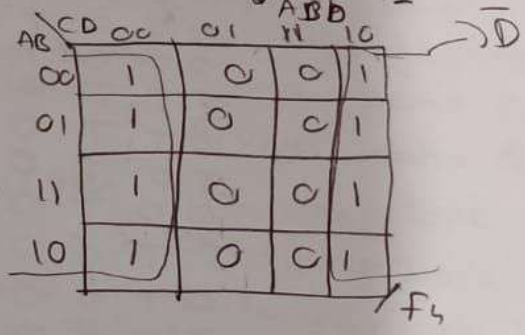
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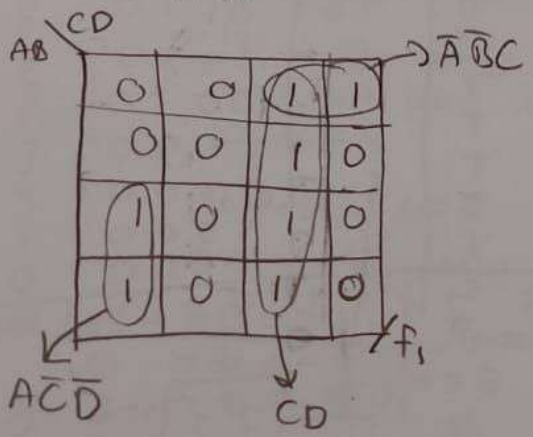




$\overline{x+y}$







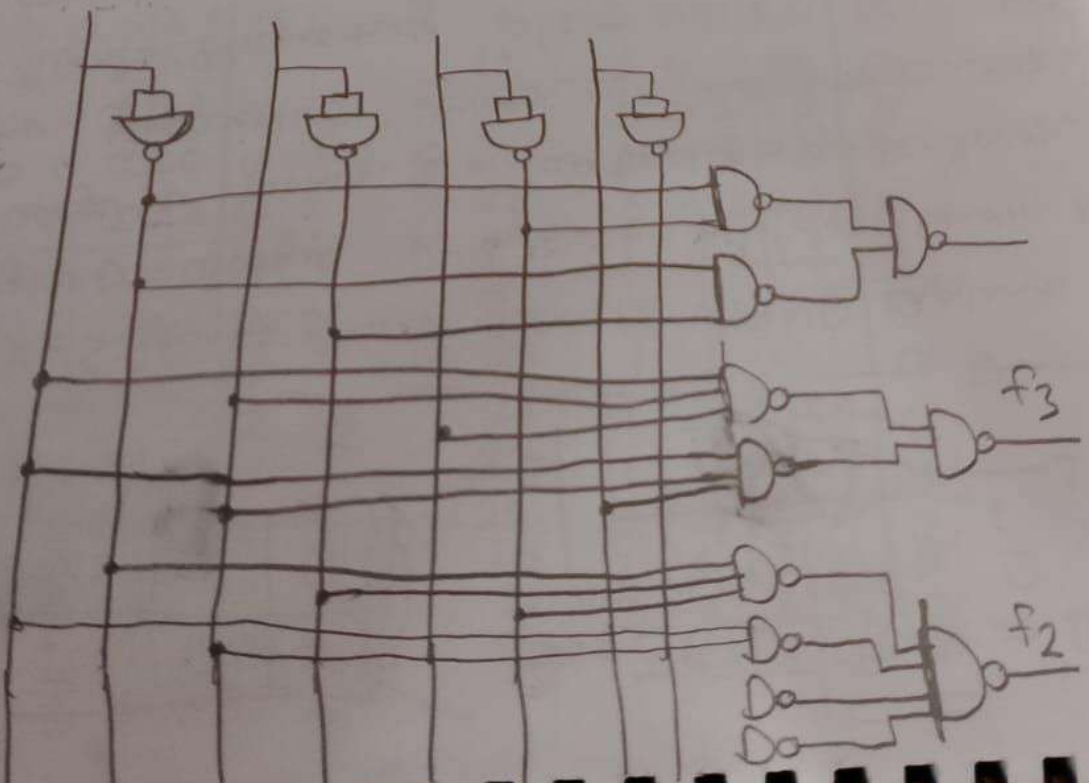
$$f_5 = \overline{\overline{A\bar{C}} + \overline{A\bar{B}}} = \overline{\overline{A}\,\bar{C}} \cdot \overline{\overline{A}\,\bar{B}}$$

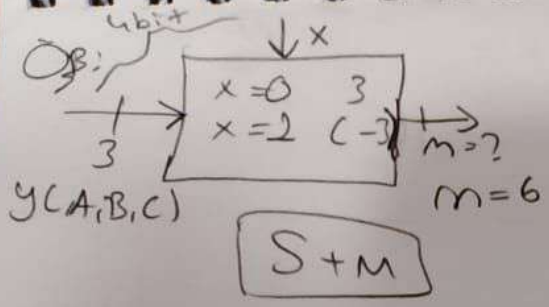
$$f_3 = AB\bar{C} + A\bar{B}D$$

$$f_2 = \bar{A}\bar{B}\bar{C} + A\bar{B}D + A\bar{B}C + AB\bar{C}\bar{D}$$

$$f_1 = A\bar{C}\bar{D} + \bar{A}\bar{B}C + CD$$

$$f_0 = \bar{D}$$



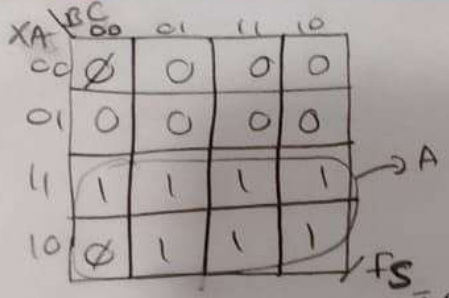


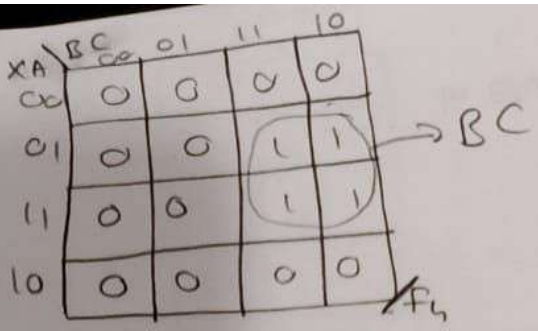
Bir lojik devrede $y(A,B,C)$ binary veri girişi, X ise kontrol girişidir. $(X=0)$ olduğu zaman $(y+3)$ ile $(X=1)$ olduğu zaman $y, (-3)$ ile çarpılmaktadır. Böyle bir devreyi yalnızca NAND kapıları kullanarak gerçekleştiriniz.

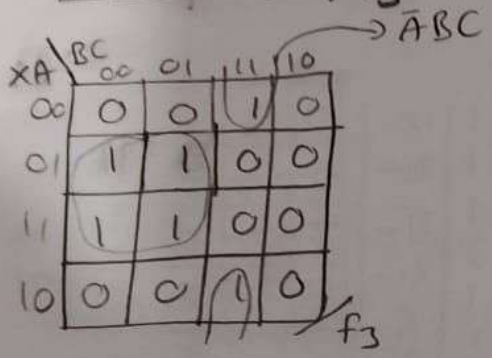
Negatif değerler için logic 1 pozitif değerler için logic 0 kullanılacaktır.

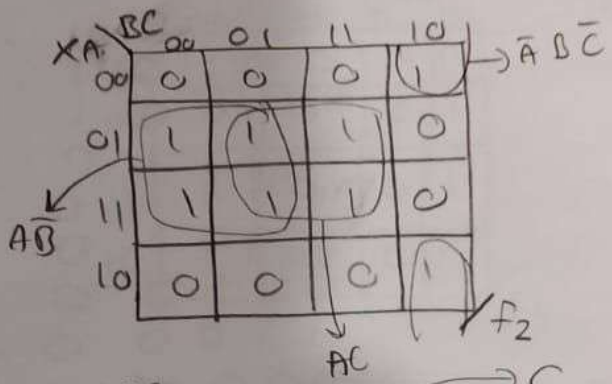
Dec.	X	y (A BC)	f	fs	f_4 f_3 f_2 f_1 f_0
0	0	000	0	ϕ	0 0 0 0 0
1	0	001	+3	0	0 0 0 1 1
2	0	010	+6	0	0 0 1 1 0
3	0	011	+9	0	0 1 0 0 1
4	0	100	+12	0	0 1 1 0 0
5	0	101	+15	0	0 1 1 1 1
6	0	110	+18	0	1 0 0 1 0
7	0	111	+21	0	1 0 1 0 1
8	1	000	0	ϕ	0 0 0 0 0
9	1	001	−3	1	0 0 0 1 1
10	1	010	−6	1	0 0 1 1 0
11	1	011	−9	1	0 1 0 0 1
12	1	100	−12	1	0 1 1 0 0
13	1	101	−15	1	0 1 1 1 1
14	1	110	−18	1	1 0 0 1 0
15	1	111	−21	1	1 0 1 0 1

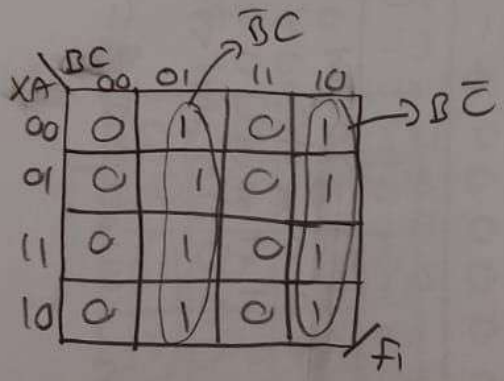
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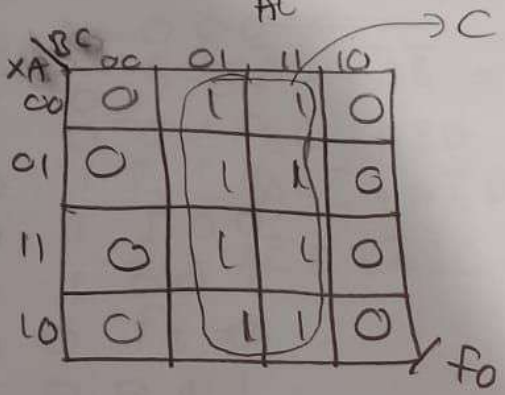












$f_S = X$

$f_4 = AB + \bar{A}BC$

$f_3 = A\bar{B} + \bar{A}BC$

$f_2 = A\bar{B} + AC + \bar{A}BC$

$f_1 = \bar{B}C + B\bar{C}$ $\quad f_1 = \bar{B}C + B\bar{C} = B \oplus C$

$f_0 = C$

6. hafta